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UNIT-5

- ① Antidiabetic agents
- ② local anesthetic

chapter-1 } → Antidiabetic agents

ImpQues

- i) classify oral hypoglycemic drugs. Write MOA of sulfonyl ureas with the synthesis of Tolbutamide.
- ii) Classify Sulfonyl ureas and Biguanides and write their mechanism.
- iii) write a short note on Glucosidase inhibitors.
- iv) Define & classify local anaesthetic & write SAR, synthesis & MOA of -
 - Benzocaine
 - Procaine
 - Articaine

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Chapter-1 Unit-5th Antidiabetic agents

- These are those agents which are used in the treatment of diabetes mellitus.
- These agents are used to lower the blood sugar level in patients suffering from hyperglycemia.

Also known as

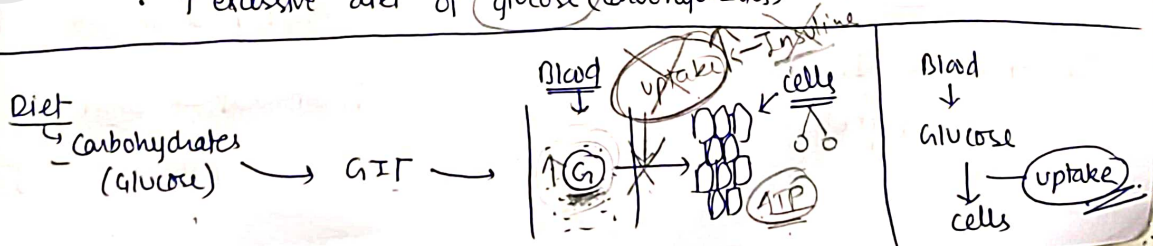
- Anti-hyperglycaemic agents
- hypoglycaemic agent
or sometimes Oral hypoglycaemic agents

• Diabetes mellitus →

- It is a metabolic ^{disease} ↑ in which there are high blood sugar levels (hyperglycemia) over a prolonged period.
- In which conditions a person's body cannot control the level of sugar in the blood.

Causes →

- defects in insulin formation, secretion and action.
- Obesity and an inactive lifestyle
- family history
- ↑ excessive diet of glucose (carbohydrates)



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Symptoms →

- polydipsia (excess thirst)
- polyphagia (excess hunger)
- polyuria (excess urination)
- Glycosuria (glucose in urine)
- lethargy, fatigue

Types of DM

Type-I diabetes
mellitus

IDDM - Insulin dependent
diabetes mellitus

- A chronic condition in which the pancreas produce little or no insulin.

- mostly occurs in children
- pancreatic β -cells are destroyed

Treatment → Insulin
injection.

Type-II diabetes
mellitus

NIDDM - Non-Insulin dependent
diabetes mellitus

- A chronic condition that affect the way the body processes blood sugar (glucose)

↓ Insulin resistance
↓ Glucose uptake by the cells.

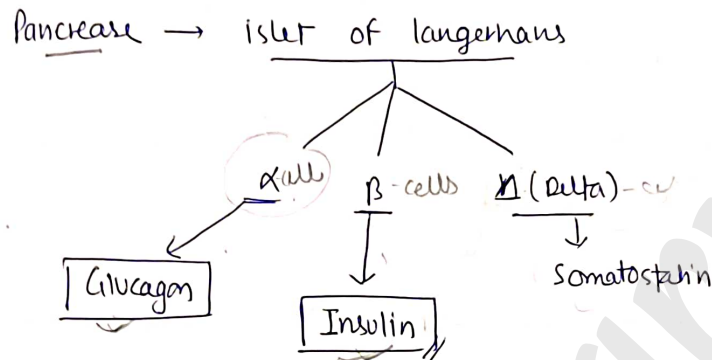
- mostly occurs in adults.

Treatment → Oral
hypoglycemic agents

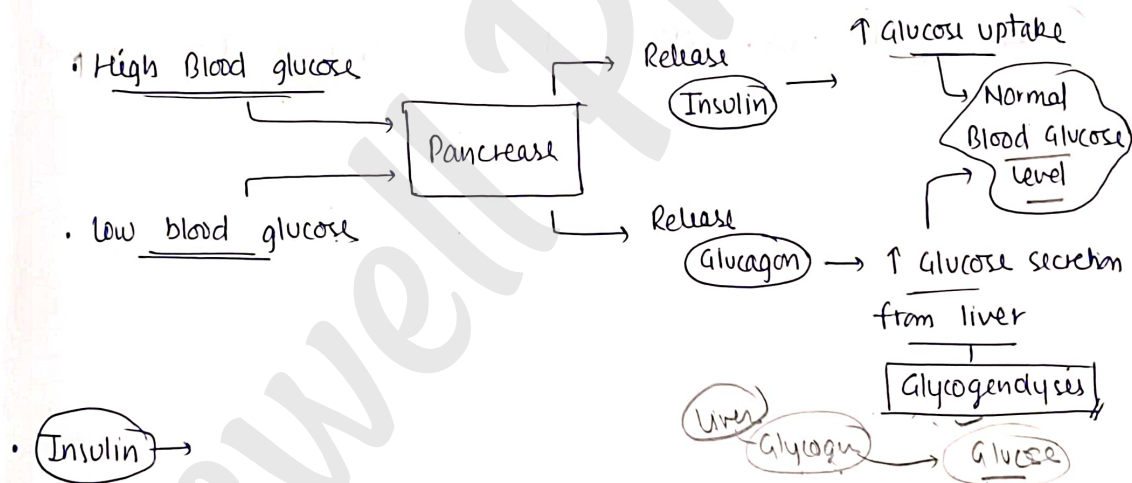
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• Insulin - Insulin + its preparations.



Regulation of Blood sugar (glucose) level - normally

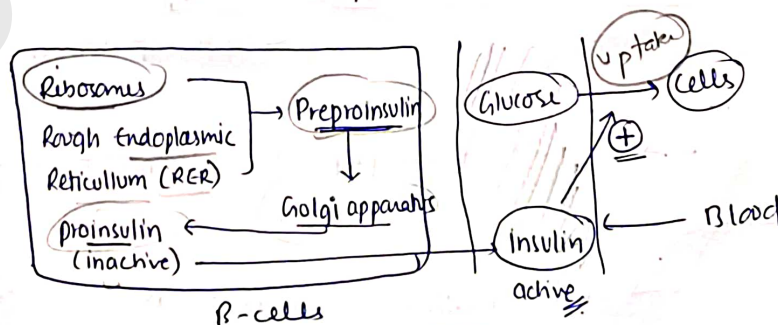


• Insulin →

- It is peptide cells secreted by β -cells of pancreas
- Discovered by Banting and Best in 1921 & used in 1922.

Synthesis →

It is synthesized in pancreatic β -cells.



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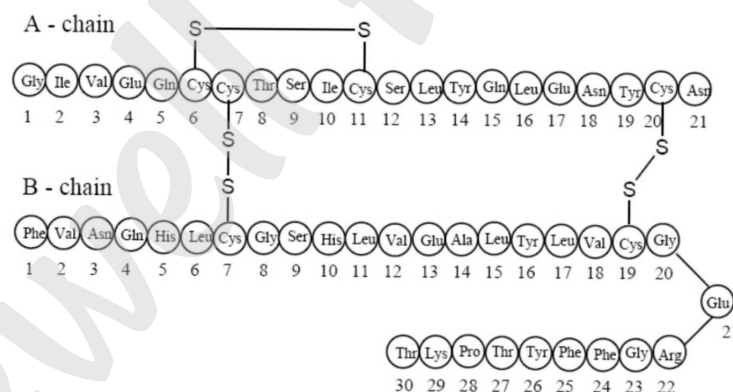
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- firstly through RER, preproinsulin is produced which further stored in golgi apparatus in the form of proinsulin (inactive form).

• Now, this proinsulin is secreted into blood in the active form Insulin.

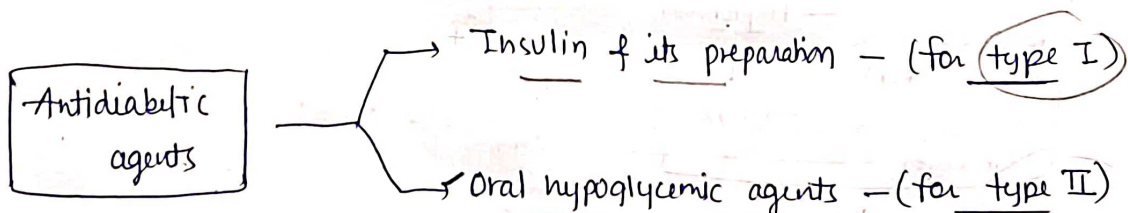
* Now, this Insulin is composed of 2 chains (A+B).

(A) chain has 21 amino acid residue, whereas (B) chain has 30 amino acid residue and both chain are attached to each other by 2 disulphide (-S-S-) bond



structure of insulin

• Now, synthetic insulin is prepared through Bovine, Porcine, Recombinant human insulin
 source of insulin.



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① Insulin preparations — (injection) 📌 CLASSIFICATION 📌

Short acting	Intermediate acting	Long acting
Regular	Isophane (NPH)	Protamine zinc
Lispro	Lente	Ultra Lente
Insulin zinc	Biphasic insulin aspart	Insulin glargine
Insulin aspart		Insulin detemir
		Insulin degludec

Regular insulin: Also called as neutral or soluble insulin. Rapid acting with 0.5-1 hr duration of action.

Lispro insulin: Rapid acting with 6-8 hr duration of action. In the carboxyl terminal of B-chain, lysine and proline residues are reversed. It is a recombinant human insulin.

Insulin Zinc: It is a suspension of insulin and zinc chloride, having 6-8 hr duration of action.

Insulin aspart: Synthetic form of human insulin where a single amino acid, proline (B-28) is replaced by aspartic acid. It has 3-5 hr duration of action.

Isophane insulin: Also called as Neutral Protamine Hagedorn (NPH) insulin. It is intermediate acting with 18-24 hr duration of action. It is composed of zinc, protamine, and regular insulin.

Lente insulin: It is intermediate acting with 18-24 hr duration of action. It is composed of acetate buffer, zinc, and regular insulin.

Biphasic insulin aspart: It is intermediate acting with 24 hr duration of action. It is composed of 30% soluble insulin aspart and 70% protamine bound insulin aspart.

Protamine zinc insulin: It is composed of insulin, zinc chloride, and protamine. It has duration of action upto 36 hrs.

Ultra lente insulin: It contains 65% of lente insulin, having duration of action upto 36 hrs.

Insulin glargine: It is obtained by addition of two arginine residues in B-chain carboxy terminal and by replacement of asparagine with glycine in A-21 position of human insulin. It has duration of action upto 24 hrs.

Insulin detemir: It is obtained by addition of a fatty acid (myristic acid) to lysine residue in B-29 position of human insulin. It has duration of action upto 24 hrs.

Insulin degludec: It is obtained by addition of hexadecanedioic acid to lysine residue in B-29 position of human insulin. It has duration of action upto 24 hrs.

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② Oral hypoglycemic agents →

These are those agents which are used for type-II DM and are used orally.

• Classification →

i) Sulfonyl ureas → Tolbutamide*, chlorpropamide, Glipizide, Glimepiride

ii) Biguanides → Metformin

iii) Thiazolidinediones → Pioglitazone, Rosiglitazone

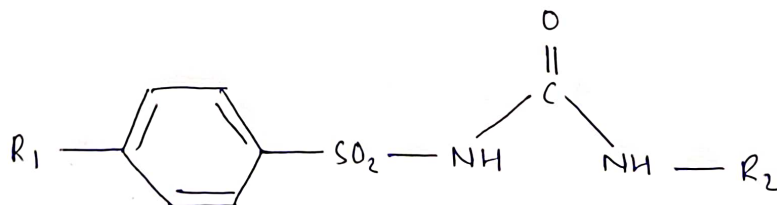
iv) Meglitinides → Repaglinide, Nateglinide

v) Glucoside inhibitors → ~~Acarbose~~, Voglibose, Acarbose

i) Sulfonyl ureas →

first class of oral hypoglycemic drugs and used for the treatment of Type-II diabetes mellitus.

• General formula →



• SAR →

• (R₁) - Add -CH₃, -COCH₃, -NH₂, -Cl, -Br, -CF₃ etc. enhance the

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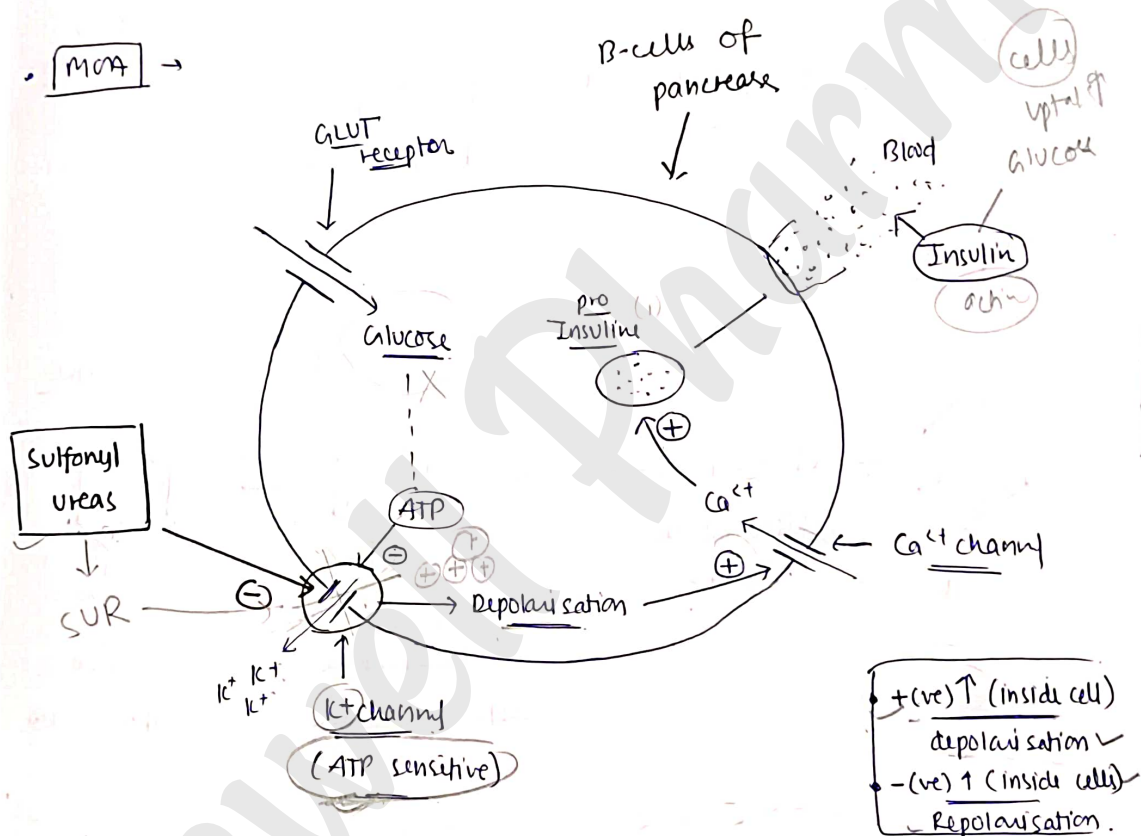
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antihyperglycemic activity

eg. chlorpropamide > tolbutamide

Also write some essential for activity.



- Sulfonyl ureas binds to the sulfonyl ureas receptor (SUR) present in pancreatic-β cells.
- Then, it leads to closure of K⁺ channel (ATP sensitive) and depolarise the beta cells.
- Due to depolarisation, it opens voltage gated Ca²⁺ channels and stimulates β-cells to secrete more insulin.

Drugs → Tolbutamide, chlorpropamide, Glipizide, Glimipizide

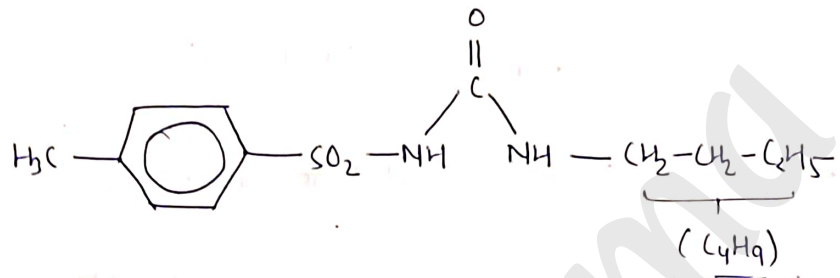
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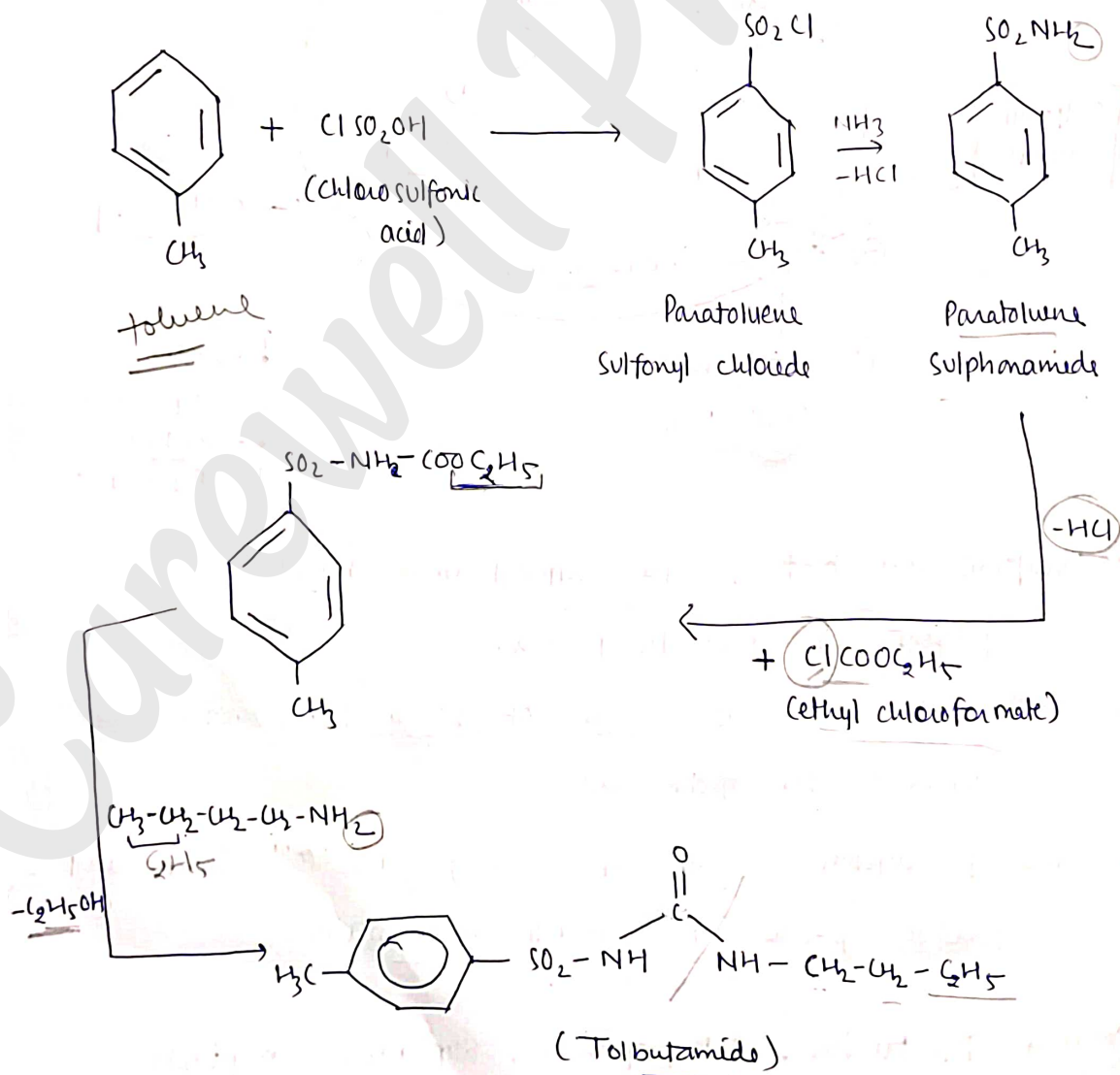
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⑤ Tolbutamide →



- It is least potent and short acting sulfonyl ureas.

Synthesis →



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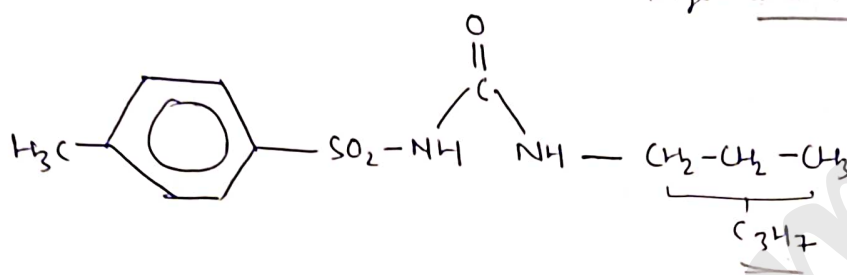
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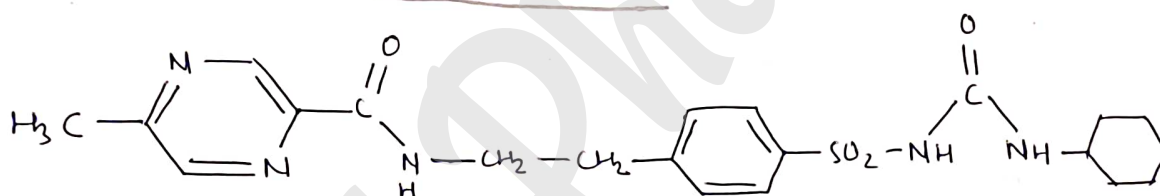
① Chlorpropamide →

— longer duration of action.



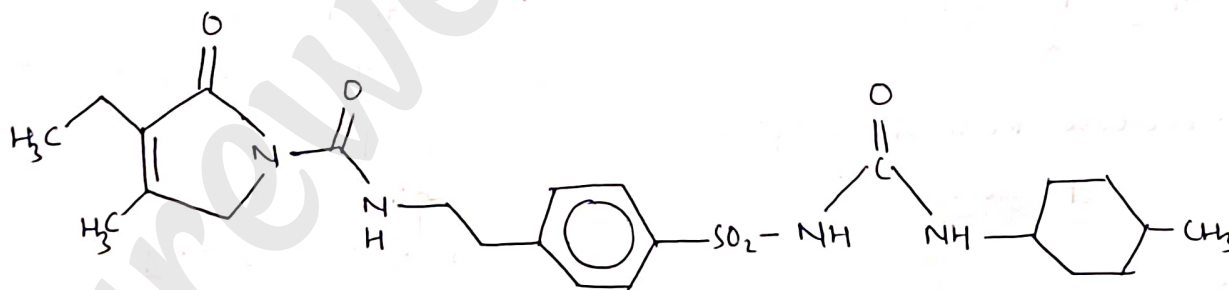
② Glipizide →

• It has quick onset of action.



③ Glimipinide →

• long acting potent sulfonylurea.



Uses →

all of ~~them~~ almost same uses →

• They are used in the treatment of Type-II DM.

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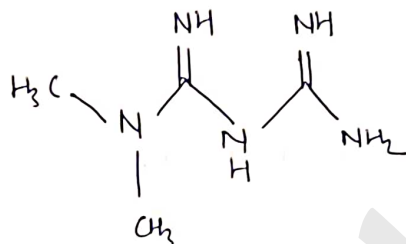
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11) Biguanides -

metformin

- Also known as Insulin sensitizers.



MOA →

metformin

other mechanism
↓

- Decrease intestinal absorption of glucose
- Increase glucose uptake and their utilization.

uses →

- used to treat type-II diabetes mellitus.

- Also used for

polycystic ovarian syndrome

Liver

Activate AMPK (Adenosine mono phosphate kinase)
enzyme

↓
Gluconeogenesis

Pyruvate
Lactate

⊖

Glucose
produce

↓
reduce

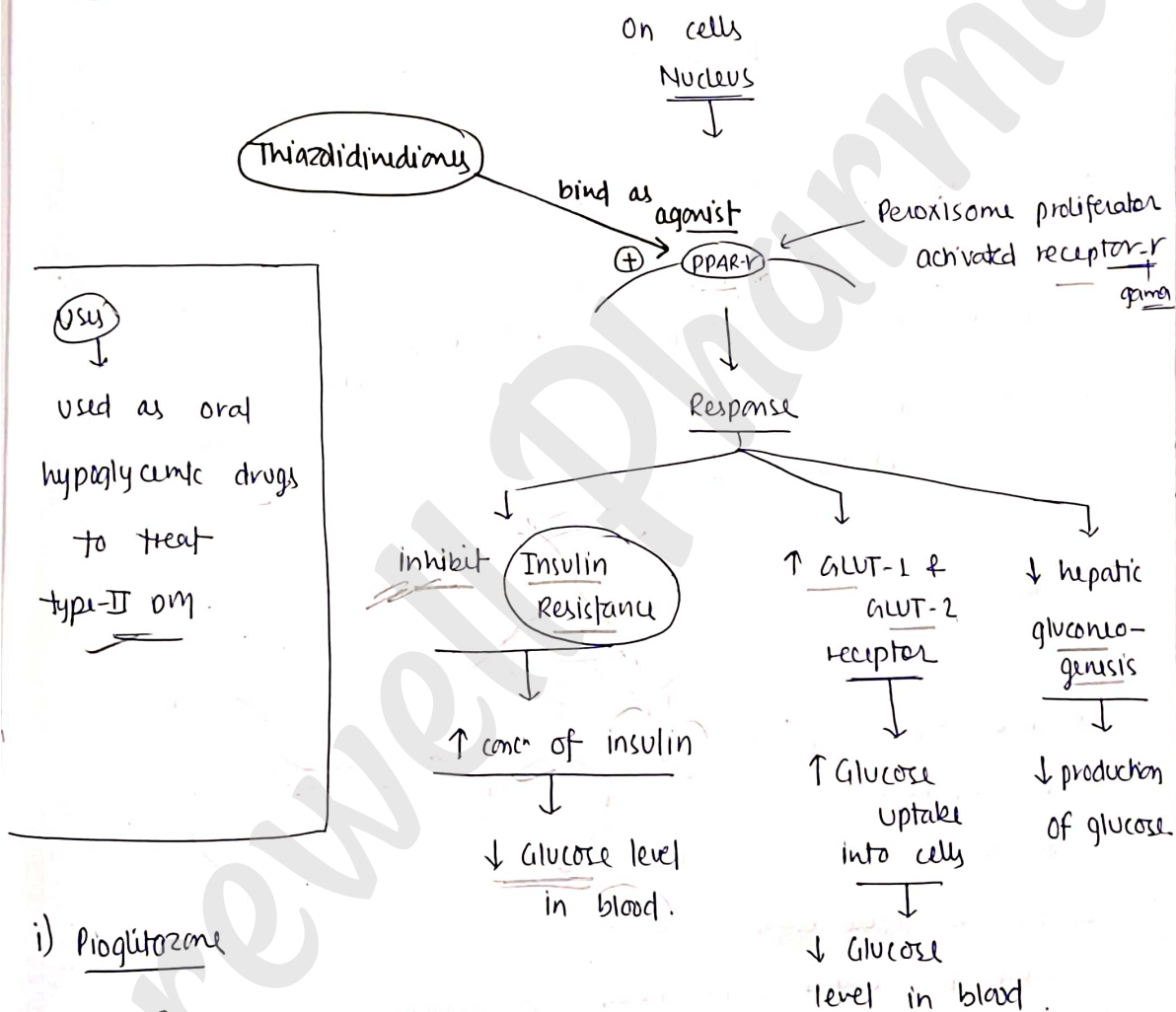
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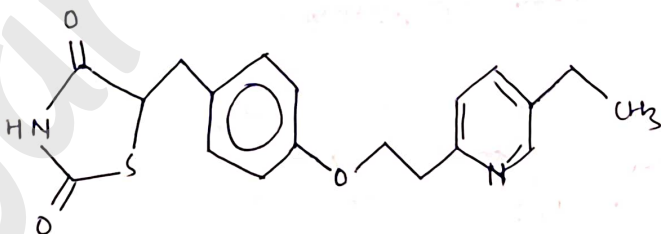
iii) Thiazolidinediones →

Drugs → Pioglitazone, Rosiglitazone

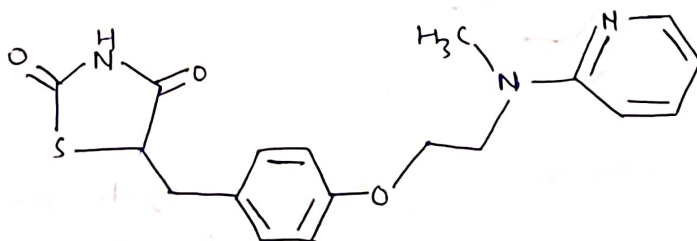
(MOA) →



i) Pioglitazone



ii) Rosiglitazone



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iv) Meglitinids →

Drugs → Repaglinide, Nateglinide

MOA →

They are used to normalize mealtime glucose excursions.

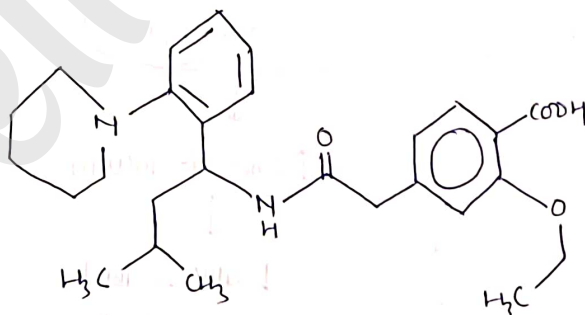
- mechanism is almost same as sulfonyl ureas but they act as analogue for binding to SUR (sulfonyl ureas receptors).
Glucose level at a time of meal
- Give their action by blocking K⁺ channel.

i) Repaglinide →

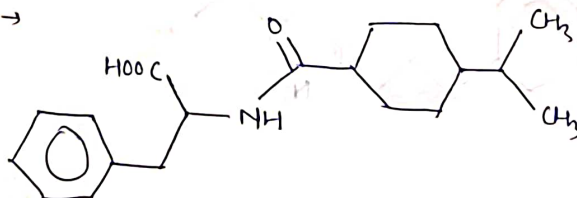
It induce fast onset and short-lasting insulin secretion.

- administered before each meal to control post-prandial hyperglycemia.

same but more potent



ii) Nateglinide →



uses for both

used as oral hypoglycemic agents for treat type-II diabetes.

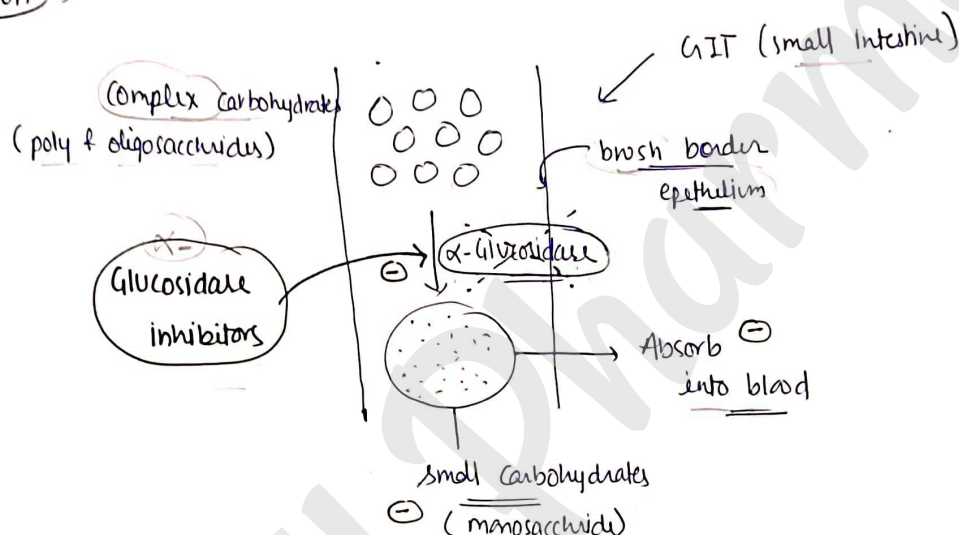
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v) Glucosidase inhibitors → (used ^{best} for Type-II diabetes)

Drugs → Acarbose, Voglibose

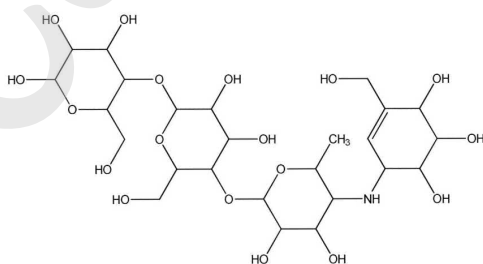
MOA →



- They decrease the breakdown of ^{complex} carbohydrates into small carbohydrates by inhibiting α -glucosidase enzyme.
- which further decrease the absorption of glucose (carbohydrate) into blood.

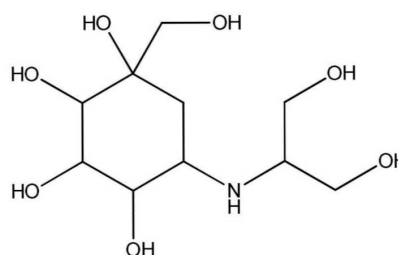
① Acarbose

- It is complex oligosaccharide which is isolated from cultures of actinophus (a bacteria)



② Voglibose →

obtained from validamycin-4



Adverse → It has side effects flatulence and loose motion.

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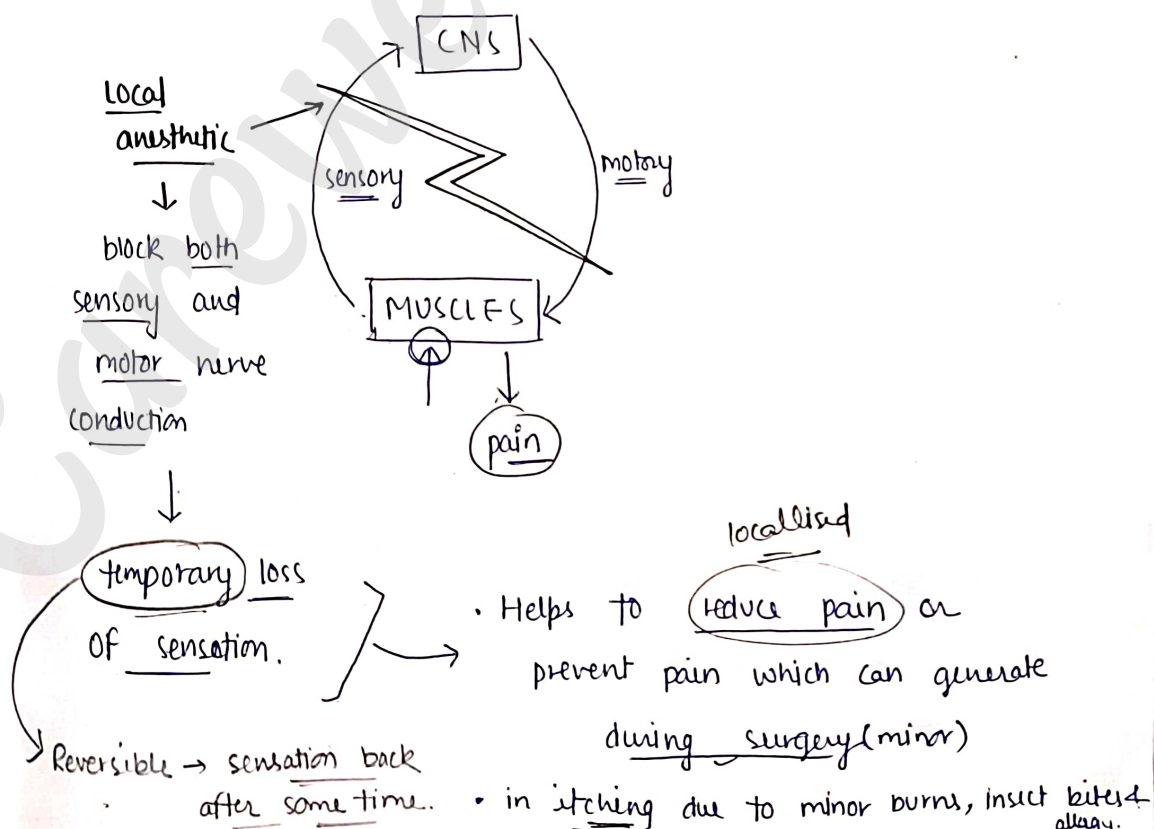
CHAPTER-2

UNIT-5

LOCAL ANESTHETICS

• local anesthetics →

- These are those agents or drugs which ~~are~~ produce reversible loss of ~~sensation~~ sensation at any particular site (area) by the neuronal conduction (action potential) on that particular sites.
- They cause muscular paralysis for some time (reversible).
- They are mainly used for minor surgery such as removal of tooth, eye surgery etc--
eg Procaine, Lidocaine



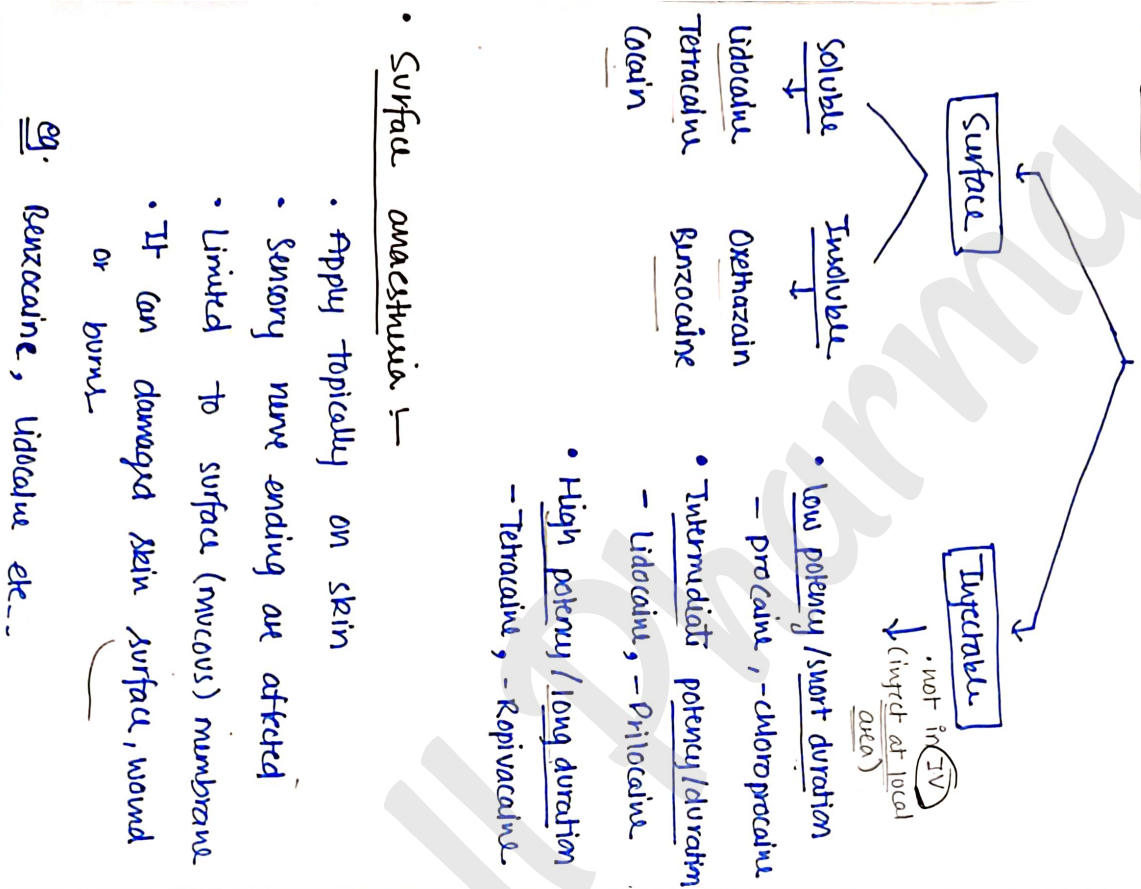
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Classification :-



• Injectable anaesthesia :-

- These drugs which inject under the skin (not through I.V.)
 - Used for minor surgery
 - Skin cut etc..
- eg. Procaine, Bupivacaine etc..

* All types of local anesthetic drugs contain tertiary amine structure. So, they are basic in nature.

- They are lipophilic in nature. so they can easily cross membrane (mostly).

- L.A. contain two types of bond

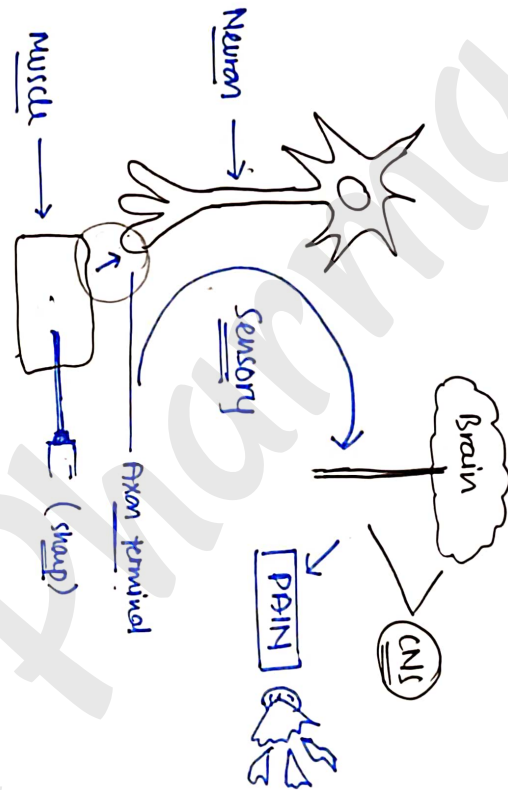
• amide bond → Lidocaine, Bupivacaine, tetracaine etc..

• Ester bond → Procaine, Cocaine, Benzocaine etc..

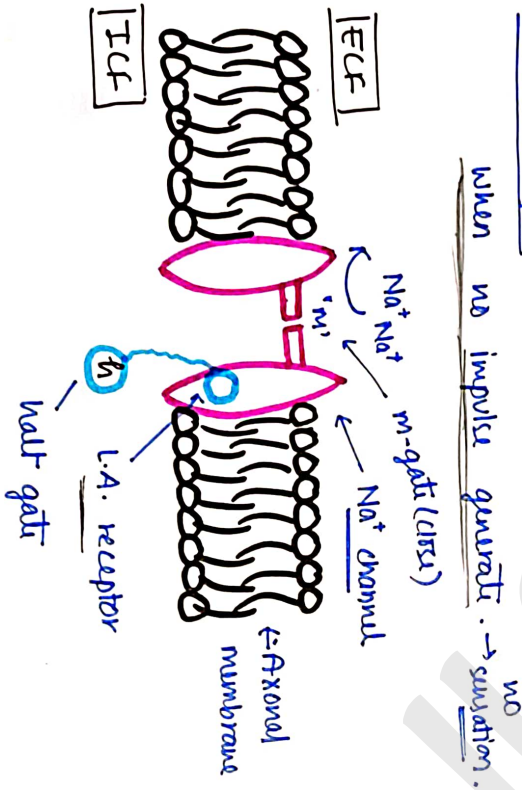
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Mechanism of Action

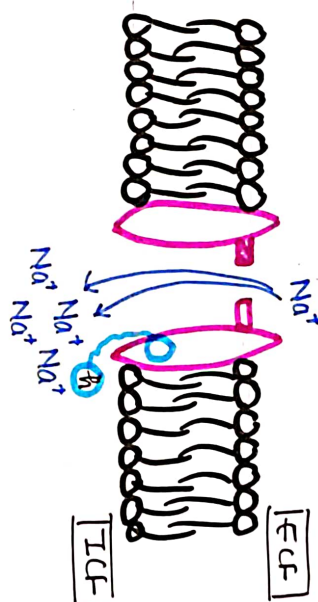


1) Normal (Resting condition) →



2) Activated →

any signal (impulse) generated → m-gate open
 Sustaining ~~Depolarisation~~ → Na⁺ ion influx
 OR
 Response.



3) Inactivated → (Blocked state).

Now, L.F. drugs introduced into body. (BH)
 ↓

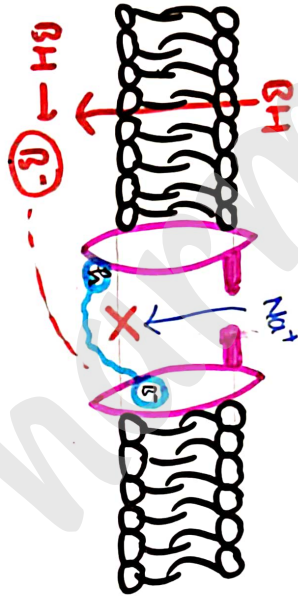
which can easily cross membrane (axonal)

Now in intracellular it dissociate into
 ionized form → BH ⇌ B⁻ + H⁺

Now, B⁻ bind with L.F. receptor with
 close (block) the Na⁺ channel with the help of

halt-(n) gati.

Depolarisation → stop → Reversibly Loss of sensation



You can write about some drugs in details.

eg. tolaine, procaine, lidocaine, benzocaine etc.

1-158

- loss of sensation (reduce pain).
- nerve block (block voltage gated Na⁺ channel).
- Used as ointment, injection etc-

Adverse effects :-

- Tongue numbness
- Muscle twitching
- Hypotension
- Redness of skin
- Asthma

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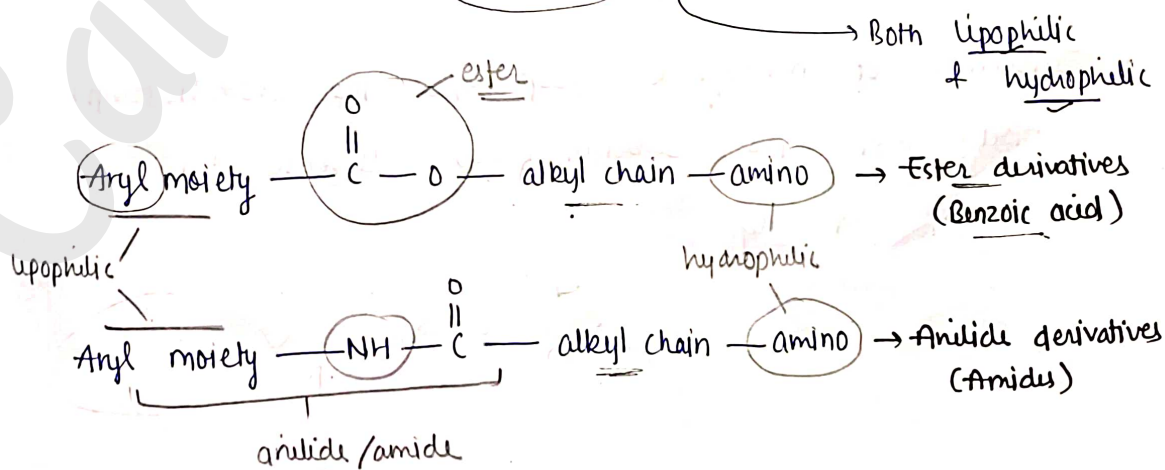
• classification →

- 1) Benzoic acid derivatives (esters) → Cocaine, Hexylcaine, Mepivacaine, Cyclomethycaine, prilocaine
- 2) Amino benzoic acid derivatives (esters) → Benzocaine*, Butamben, Procaine*, Butacaine, propoxycaine, Tetracaine, Benoxinate
- 3) Lidocaine / Anilide derivatives → lignocain (Lidocaine/Xyllocain), (amide) Mepivacaine, prilocalin, Etidocaine
- 4) Miscellaneous → Phenacaine, Diprivan, Dibucaine*

• SAR →

Structure activity relationship

Local anesthetics → Mostly are weak bases and they also have amphipilic properties.



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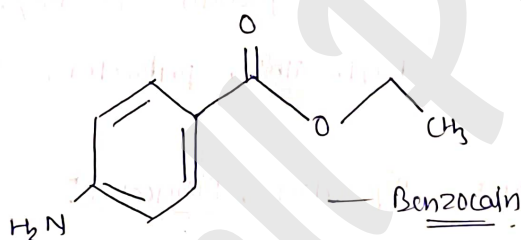
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- Substitution can be occurs at -

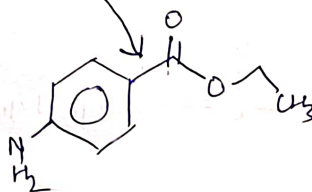
- Aryl group
- Alkyl chain
- Amino group

- Aryl groups →

- 1) Ester derivatives (Both Benzoic /amino benzoic)



- The aryl group directly attached to the carbonyl group increases the activity.
- An electron-donating group substitution at position ortho/para (both) increases the potency of local anesthetic.
- Insertion of methylene group b/w the aromatic (aryl) moiety and the carbonyl group decreases the anesthetic activity.



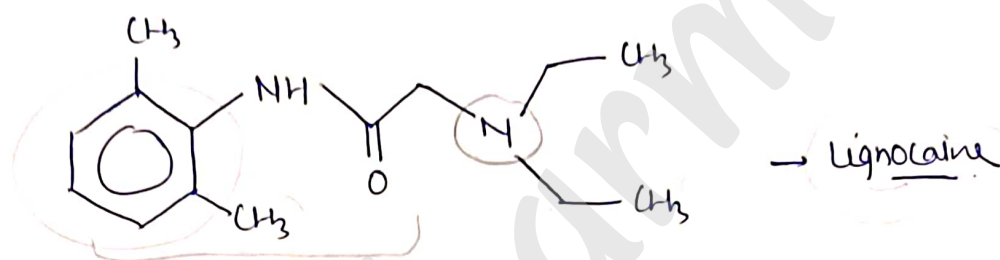
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ii) Amide /amide derivatives



- Substitution of methyl group at ortho position (both side) on aryl substitution gives longer duration of action

- Alkyl chain (for both)

- Branching in the alkyl chain especially near the ester / amide group prolonging the duration of action.

eg Prilocaine, Eticodaine



→ Amino group (for both)

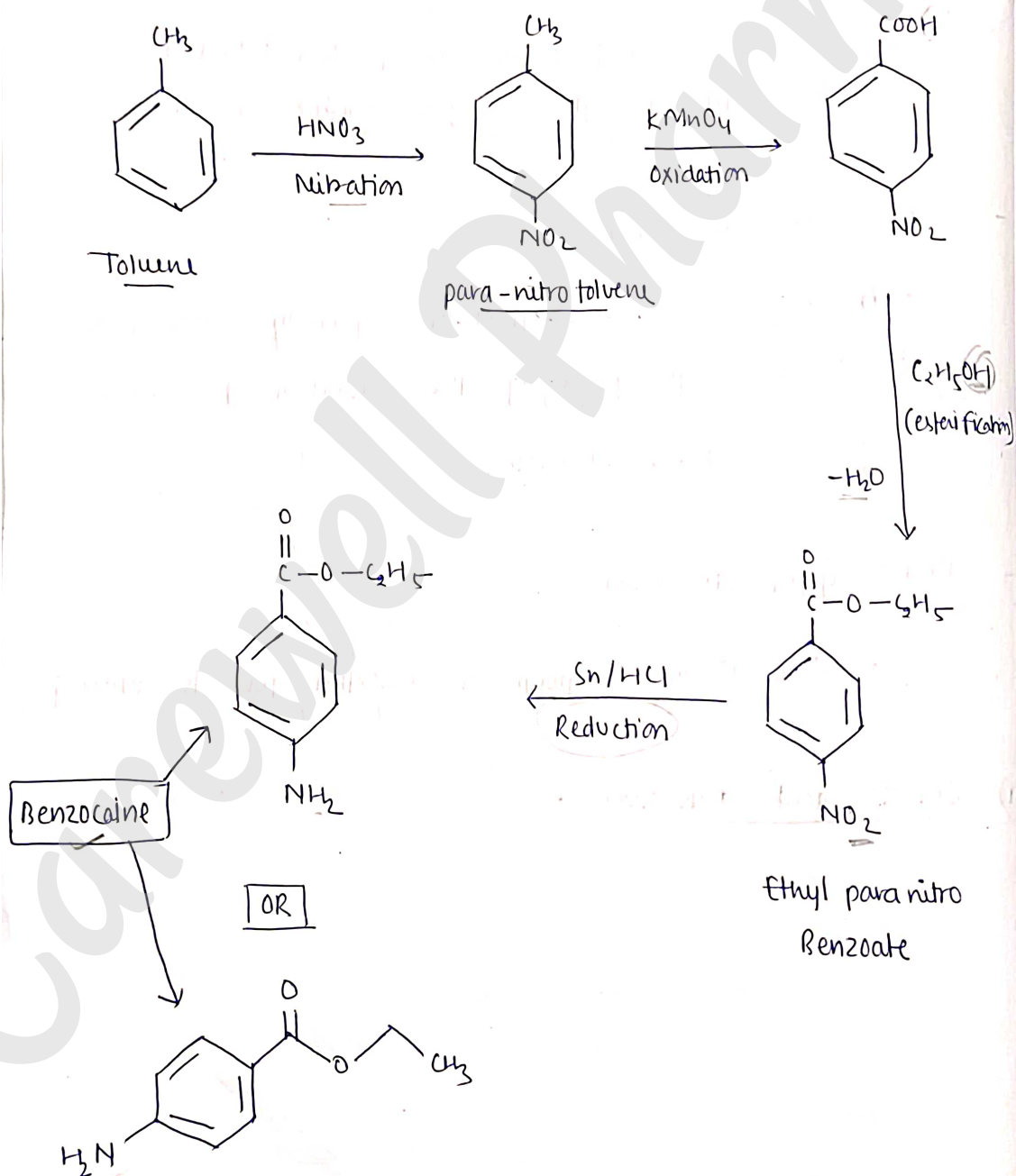
- 2° or 3° amine group is essential for its activity.

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Synthesis →

① Benzocaine →



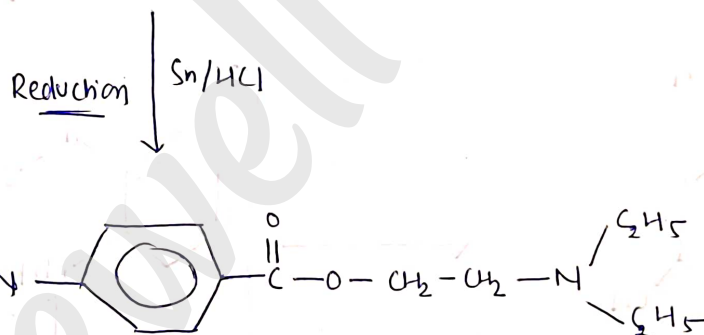
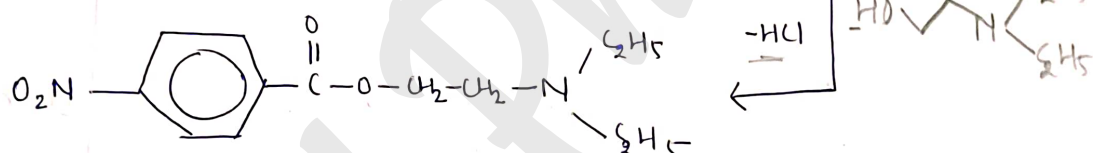
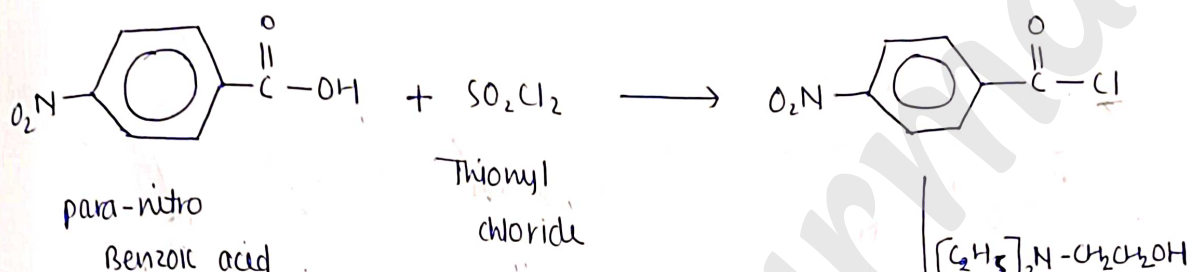
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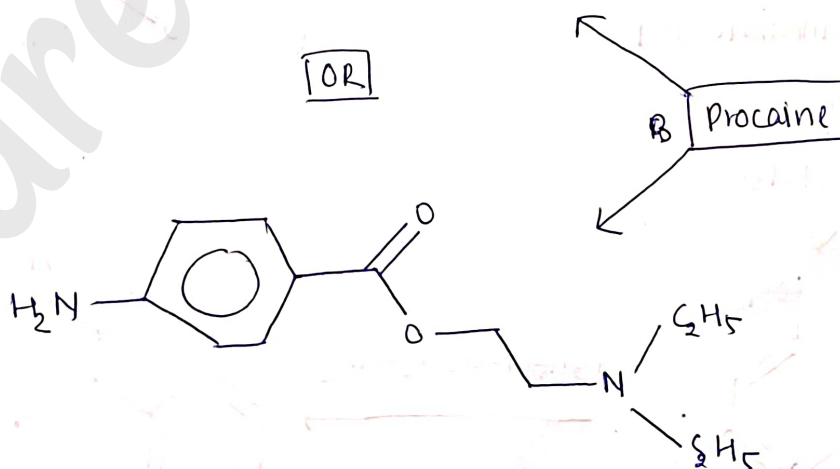
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② Procaine →



OR



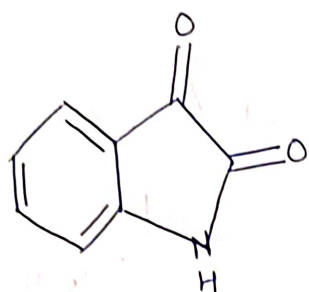
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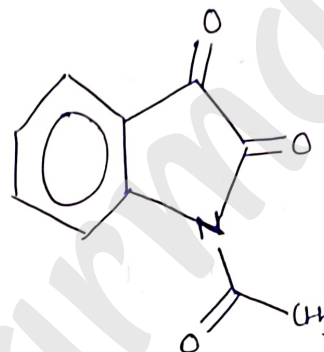
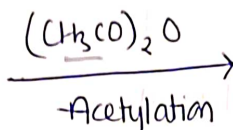
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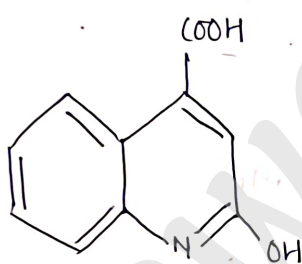
③ Dibucaine →



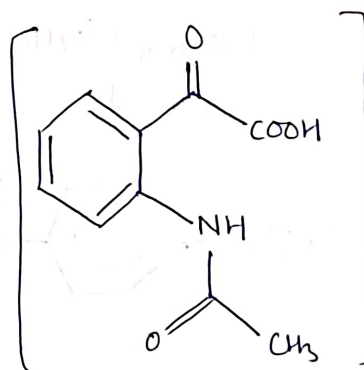
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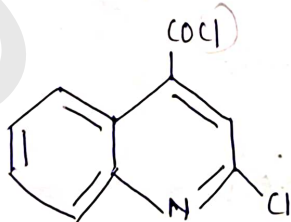
NaOH



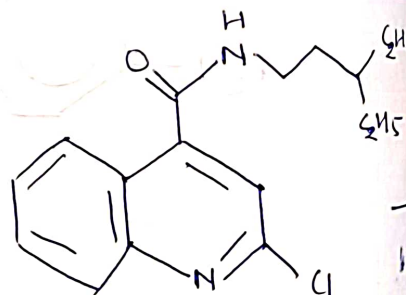
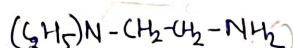
2-hydroxy cinnamic acid



PCl_5



2-chloro cinnamic
chloride



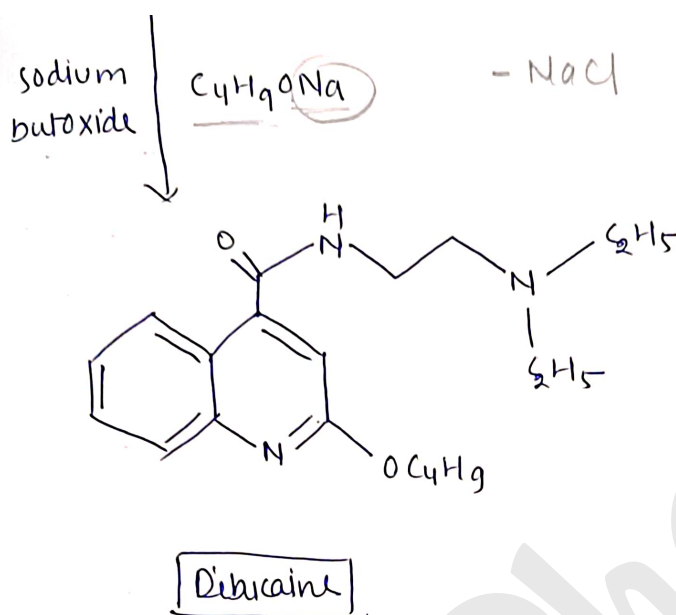
Dibucaine

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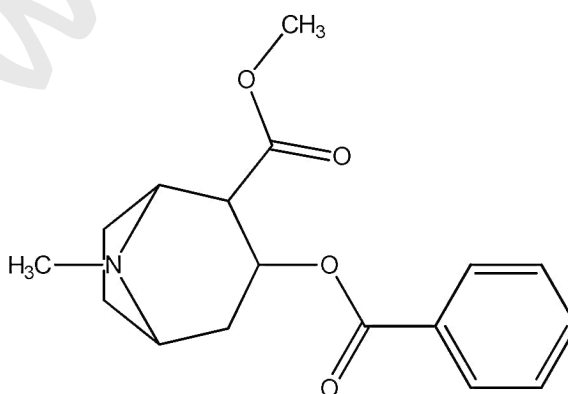
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CLASSIFICATION

1. Benzoic acid esters- Cocaine, Hexylcaine, Meprylcaine, Cyclomethycaine, Piperocaine
2. p-Aminobenzoic acid esters- Benzocaine, Butamben, Procaine, Butacaine, Propoxycaine, Tetracaine, Benoxinate
3. Amide/Anilide derivatives- Lignocaine/Lidocaine/Xylocaine, Mepivacaine, Prilocaine, Etidocaine
4. Miscellaneous agents-
Amidine derivative- Phenacaine
Carbamate derivative- Dipreron
Quinoline derivative- Dibucaine

Cocaine- 2-methoxy carbonyl-tropan-3-yl-benzoate

It is a natural alkaloid, isolated from the leaves of Erythroxylon coca. It is the first drug to be used as local anaesthetic.



Used as anaesthesia for eye, ear, nose & throat.

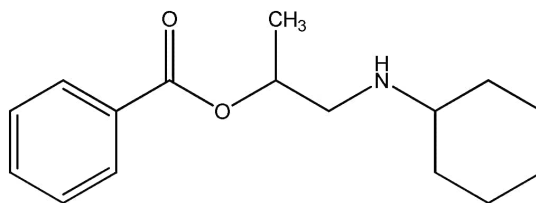
Hexylcaine- 1-cyclohexyl amino-propan-2-yl-benzoate

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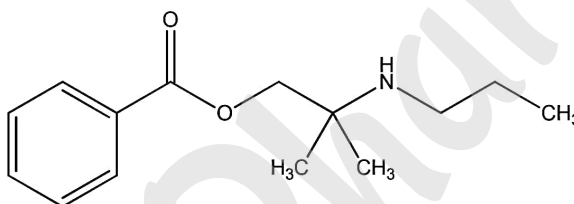
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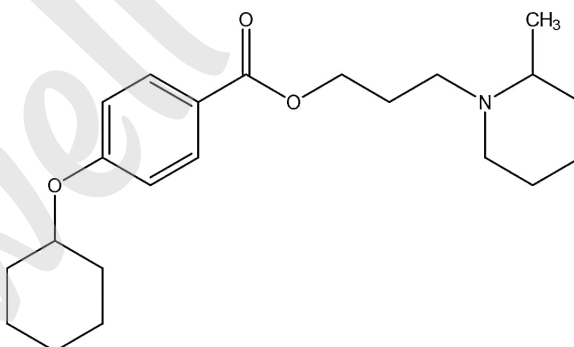
Used as surface anaesthesia.

Mepylcaine- 2-methyl-2-(propylamino) propyl benzoate



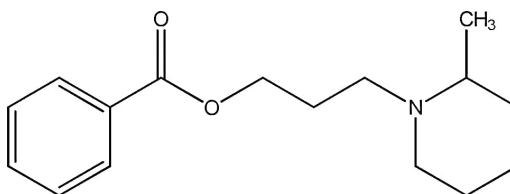
Used for dental anaesthesia.

Cyclomethycaine- 3-(2-methyl-1-piperidinyl) propyl-p-(cyclohexyloxy) benzoate



Used as topical anaesthesia.

Piperocaine- 3-(2-methyl-1-piperidinyl) propyl benzoate



Used as ocular anaesthesia.

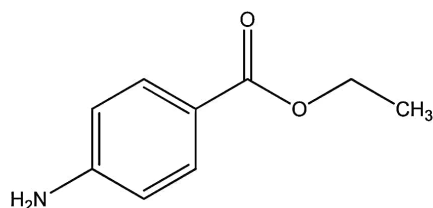
Benzocaine- Ethyl-p-aminobenzoate

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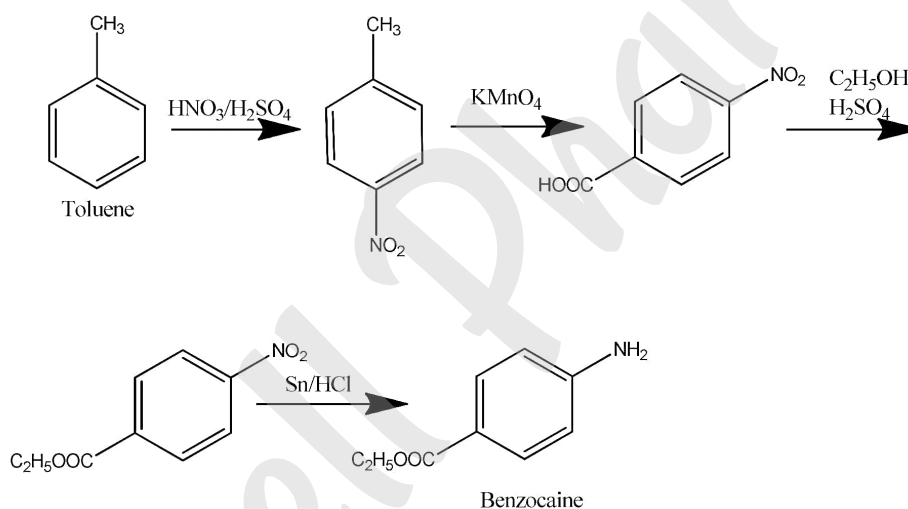
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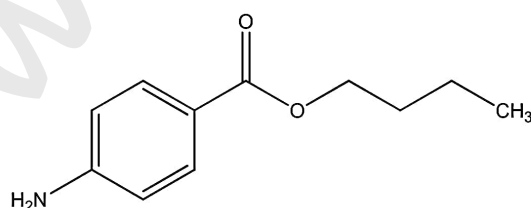


Used to reduce pain due to sunburn, teeth pain, piles, vaginal/rectal irritation.

Synthesis-



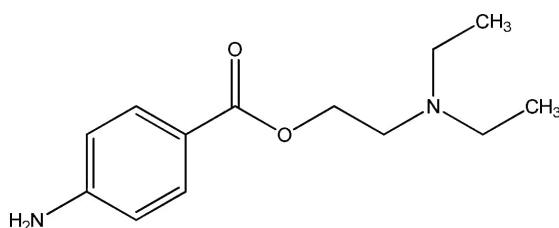
Butamben- Butyl-p-aminobenzoate



Used as surface anaesthesia for relief of pain and itching associated with anorectal disorders.

Procaine- Also called as novocaine.

2-(diethylamino) ethyl-p-amino benzoate



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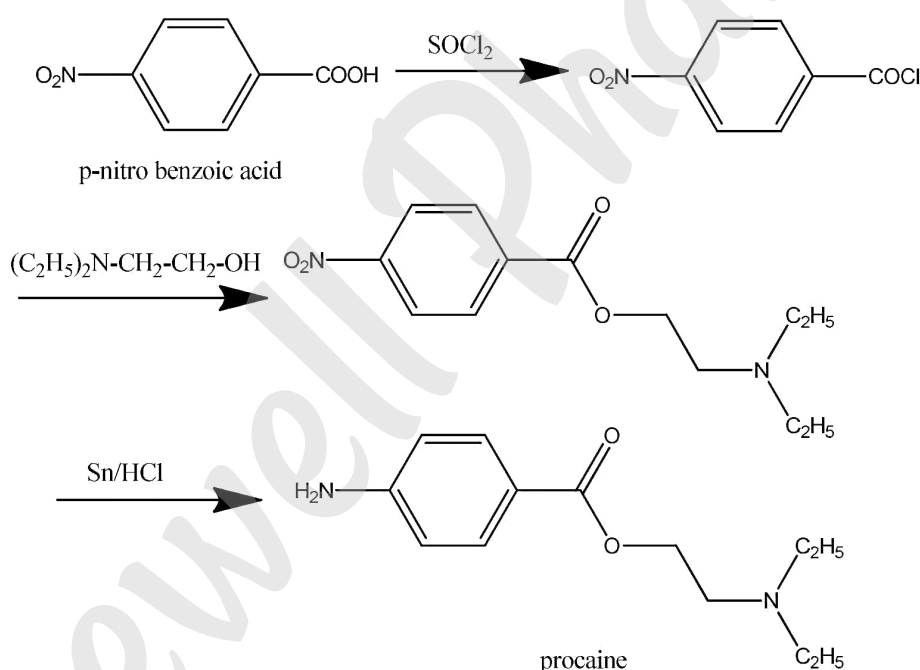
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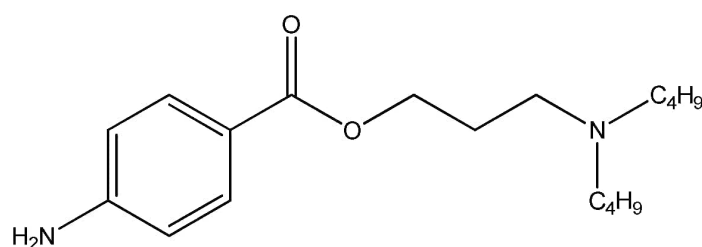
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Used as infiltration, nerve block and spinal anaesthesia. It is a vasodilator. So, adrenaline is added to it to retard the absorption and prolong its duration of action.

Synthesis-



Butacaine- 3-(dibutylamino) propyl-p-amino benzoate



Used as topical anaesthesia.

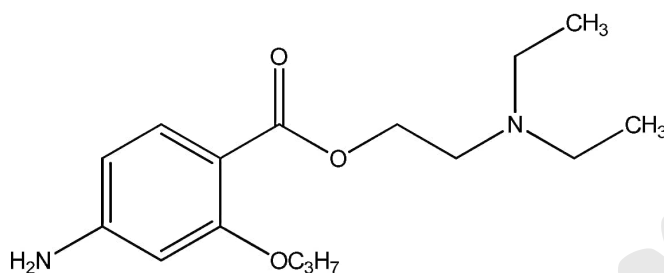
Propoxycaine- 2-(diethylamino)ethyl-2'-propoxy-4'-amino benzoate

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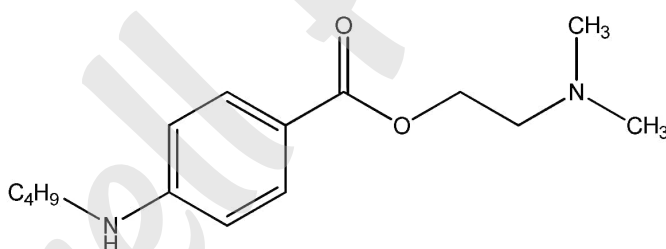
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Used as dental anaesthesia.

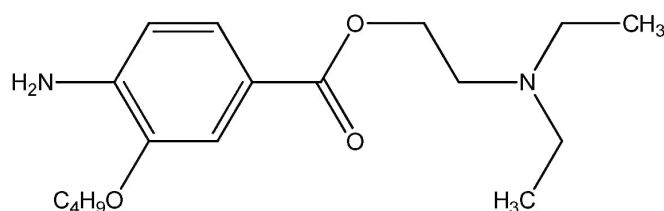
Tetracaine- 2-(dimethylamino)ethyl-p-(butylamino) benzoate



Used as topical and spinal anaesthesia.

Benoxinate- Also called as oxybuprocaine.

2-(diethylamino)ethyl-3'-butoxy-4'-amino benzoate



Used in ophthalmology.

Lignocaine- Also called as Lidocaine or Xylocaine.

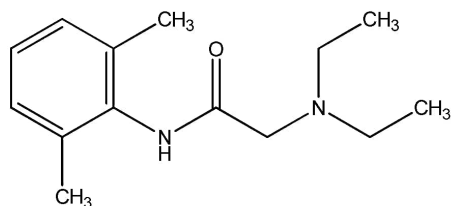
2-(diethylamino)-N-2,6-(dimethyl phenyl) acetamide

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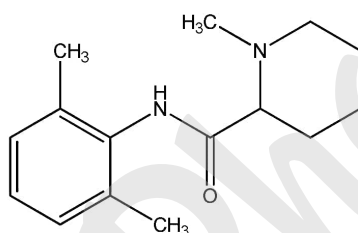
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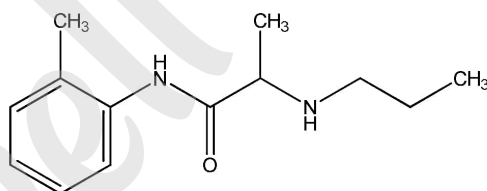
Used as infiltration, nerve block and surface anaesthesia. Also used for dental anaesthesia.

Mepivacaine- 1-methyl-N-(2,6-dimethyl phenyl)piperidin-2-carboxamide



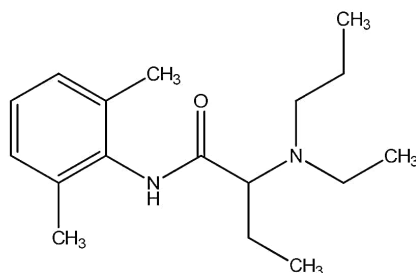
Used as epidural and nerve block anesthesia. Also used in dental procedures.

Prilocaine- N-(2-methyl phenyl)-2-propylamino propanamide



Used for infiltration, nerve block anaesthesia. A mixture of prilocaine and lignocaine (eutectic mixture) is used as topical dosage form.

Etidocaine- N-(2,6-dimethyl phenyl)-2-(ethyl propyl amino)-butanamide



Used along with adrenaline for infiltration, epidural and nerve block anaesthesia.

Phenacaine- Also known as holocaine.

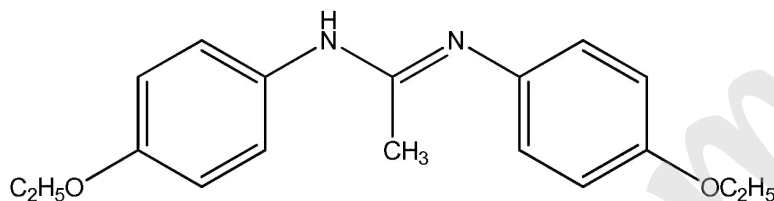
N,N'-bis(p-ethoxyphenyl)ethan imidamide

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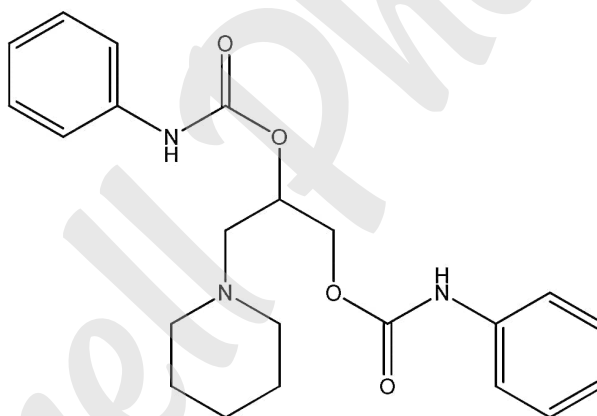
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Used in ophthalmology.

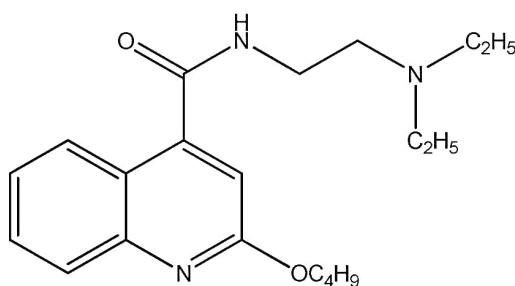
Diperoxon- 3-(1-piperidiny)-1,2-propandiyl-bis-(phenyl carbamate)



Used as topical anaesthetic.

Dibucaine- Also known as Cinchocaine. It is the most potent, most toxic, longest acting local anaesthetic.

N-(2-diethyl amino ethyl)-2-butoxy-quinolin-4-carboxamide

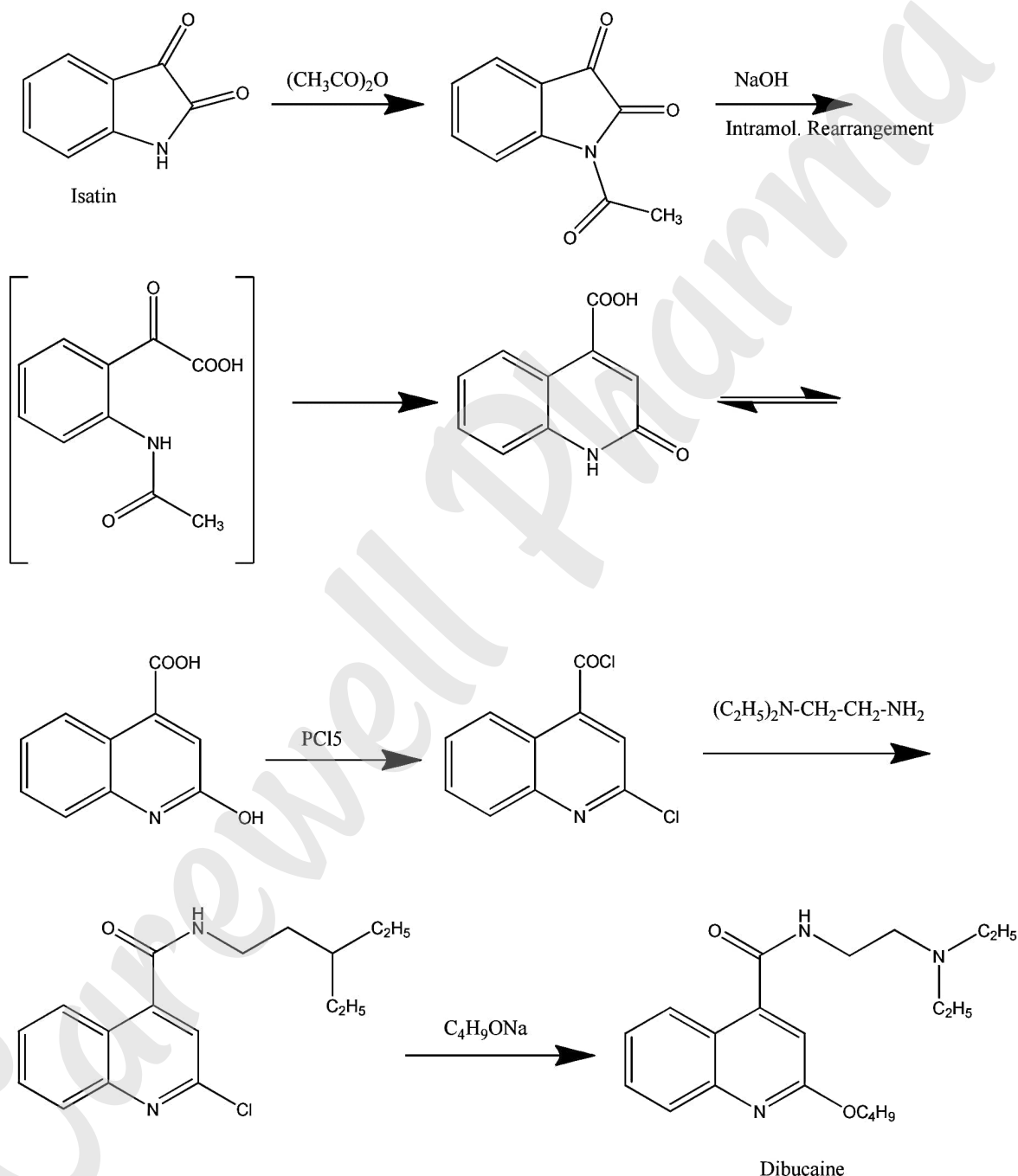


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Dibucaine is used as surface anaesthetic on less delicate mucous membrane like anal canal.

Can be used parenterally for spinal anaesthesia only.

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